

Partnership

1. Profit $\propto$ Investment $P \propto I$
2. Profit $\propto$ Time invested $P \propto T$.
3. Profit $\propto IT$.

4. If two partners,

$$\frac{P_A}{P_B} = \frac{I_A}{I_B} \times \frac{T_A}{T_B}$$

$$5. (3 \text{ part}) \Rightarrow P_A : P_B : P_C = I_A T_A : I_B T_B : I_C T_C$$

1. A & B started a business investing Rs. 4800 and Rs. 6400 respectively. But B left the business after 8 months. Find the difference between the share of profits of A & B, if the total profit at the end of the year is Rs. 12100.

Ans: 7700/-

$$\frac{P_A}{P_B} = \frac{I_A}{I_B} \times \frac{T_A}{T_B} = \frac{4800}{6400} \times \frac{12}{8} = \frac{9}{2}$$

$$P_A : P_B = 9 : 2$$

$$11P \rightarrow 12100$$

$$7P \rightarrow x$$

$$\Rightarrow \frac{1100}{12100} \times 7 \Rightarrow 7700/-$$

2. A, B, C enter into Partnership. A contributes Rs. 360 for 4 months, B contributes Rs. 420 for 3 months, C contributes Rs. 180 for 5 months. If the total profit is Rs. 400. Find the share of A in the profit.

Ans: 160/-

$$P_A : P_B : P_C = I_A T_A : I_B T_B : I_C T_C$$

$$= 360 \times 4 : 420 \times 3 : 180 \times 5$$

$$= 8 : 7 : 5$$

$$20P \rightarrow 400$$

$$8P \rightarrow x$$

$$\Rightarrow \frac{400}{20} \times 8 = \boxed{160} \Rightarrow A's \text{ share.}$$

3. A & B invest in the ratio 3:5. After 6 months C joined the business investing an amount equal to B. In what ratio they need to share the profits at the end of the year?

Ans: 6:10:5

$$I_A = 3 \quad T_A = 12$$

$$I_B = 5 \quad T_B = 12$$

$$I_C = 5 \quad T_C = 6$$

$$P_A : P_B : P_C = 3 \times 12 : 5 \times 12 : 5 \times 6$$

$$= 36 : 60 : 30$$

$$= 12 : 20 : 10$$

$$= 6 : 10 : 5$$

4. A started a business investing Rs. 25000. 6 months later B joined him by investing Rs. 15000. After another 6 months B invested an additional amount of Rs. 15000. At the end of 3 years they earned a profit of 2,47,000/-. Find the share of B in the profit.

Ans:- 1,17,000/-

$$P_A : P_B = 25 \times 36 : 15 \times 30 + 15 \times 24$$

$$= 10 : 9$$

B's share $\Rightarrow$

$$19P \rightarrow 2,47,000$$

$$9P \rightarrow x \quad \frac{2,47,000}{19} \times 9 \Rightarrow 1,17,000$$

5. A & B started a business with investments in ratio 5:3. After 6 months C joined them with investment ratio between B and C as 2:3. If the annual profit earned by them was Rs. 12,300. Find the difference b/w shares of B & C in the profit.

Ans: 900/-

$$A : B : C$$

$$5 : 3 : 3$$

$$2 : 2 : 3$$

$$\hline 10 : 6 : 9$$

$$P_A : P_B : P_C = 10 \times 12 : 6 \times 12 : 9 \times 6$$

$$= 20 : 12 : 9$$

$$41P \rightarrow 12,300$$

$$3P \rightarrow \frac{12,300}{41} \times 3 = 900/-$$

6) A started a business with a capital of Rs. 54000 and admitted B & C after 4 months and 6 months respectively. At the end of the year, the profit was divided in the ratio 1:4:5. Find the difference b/w the capitals invested by B and C.

Ans: 2,16,000/-

$$1:4:5 = 54000 \times 12 : B \times 8 : C \times 6$$

$$\frac{1}{4} = \frac{54000 \times 12^{\frac{1}{2}}}{B \times 8^{\frac{1}{2}}}$$

$$\frac{1}{5} = \frac{54000 \times 12^{\frac{1}{2}}}{C \times 6}$$

$$B = 54000 \times \frac{12}{8}$$

$$C = 54000$$

$$B = 81000$$

$$\Rightarrow B - C = 216000/-$$

7. A, B, C started a business, A invests for 4 months and claims 1/8th of total profit, B invest for 6 months and claims 1/3rd of total profit. while C invests Rs. 1560 for 8 months and how much does A & B invests together.

Ans: 2000/-

$$P_A : P_B : P_C = 3 : 8 : 13$$

$$8:13 \Rightarrow 24$$

$$A \times 4 : B \times 6 : 1560 \times 8 = 3 : 8 : 13$$

$$\frac{A \times 4}{1560 \times 8} = \frac{3}{13}$$

$$A = 360 \times 2 = 720$$

$$\frac{B \times 6}{1560 \times 8} = \frac{8}{13}$$

$$B = 1280$$

$$A + B = 2000$$

8. A, B, C started business. A invests $33\frac{1}{3}\%$ of total capital. B invest 25% of remaining capital, C invests the rest. If total profit at end of the year is Rs. 1,62,000. Find the share of A in the profit.

Ans: 54000/-

$$P_A : P_B : P_C = I_A : I_B : I_C$$

$$A \rightarrow \frac{1}{3}$$

$$B \rightarrow \frac{1}{4} \left(\frac{2}{3} \right) = \frac{1}{6}$$

$$C \rightarrow 1 - \left[\frac{1}{3} + \frac{1}{6} \right] = \frac{1}{2}$$

$$P_A : P_B : P_C = \left[\frac{1}{3} : \frac{1}{6} : \frac{1}{2} \right] \times 6$$

$$= 2 : 1 : 3$$

A's share,

$$= \frac{2}{6} \times 162000$$

$$= 54000/-$$

9. A, B, C started a business with their capitals in the ratio 2:3:5. A increased the capital by 50% after 4 months. B increased his capital by $33\frac{1}{3}\%$ after 6 months. C ~~in~~ withdraw 50% of his capital 8 months. If the total profit at end of year was Rs. 1,26,000. Find the difference b/w shares of A & C in the profit.

Ans: 12600/-

$$PA:PB:PC = 2 \times 4 + 3 \times 8 : 3 \times 6 + 4 \times 6 : 5 \times 8 + 2.5 \times 4$$

$$= 32 : 42 : 50 = 16 : 21 : 25$$

$$62P \rightarrow 26800$$

$$C-A \quad 9P \rightarrow \frac{\begin{array}{r} 74 \\ 43400 \\ \hline 86800 \\ 62 \\ \hline 31 \end{array}}{31} \times 9 = 12600/-$$

$$\begin{array}{r} 31 \times 4 \\ 14 \\ \hline 31 \overline{) 434} \\ 31 \\ \hline 124 \\ 124 \\ \hline 0 \end{array}$$

10) A, B and C started a business. Each investing Rs. 20000. After 5 months A withdraw Rs. 5000, B withdraw Rs. 4000. And C invest Rs. 6000 more. At the end of the year. A total profit of Rs. 69,900 is achieved. Find the share of A in the profit.

$$\boxed{\text{Ans:- } 20,500/-}$$

$$PA:PB:PC = 20 \times 5 + 15 \times 7 : 20 \times 5 + 16 \times 7 : 20 \times 5 + 26 \times 7$$

$$= 100 + 15 \times 7 : 100 + 16 \times 7 : 100 + 26 \times 7$$

$$= 205 : 310 : 282$$

$$\begin{array}{r} 3 \\ 15 \times 7 \\ 105 \\ 105 \\ \hline 210 \\ 26 \times 7 \\ 182 \end{array}$$

$$699P \rightarrow 69900$$

$$205P \rightarrow \frac{\begin{array}{r} 100 \\ 69900 \\ \hline 699 \end{array}}{699} \times 205 = 20500/-$$

11) Two businessmen A & B invest in a business in the ratio 5:8. They decided to reinvest 30% of the profit, they earned back into the business. The remaining profit was distributed among themselves. If A's share of profit is ₹7500. Find the total

profit made by business.

Ans: 3,25,000/-

$$A's \text{ share} = 87500$$

$$\frac{5}{13} \times \frac{70}{100} \times \text{total profit} = 87500$$

$$\begin{aligned} \text{Total profit} &= \frac{12500}{87500} \times \frac{100}{70} \times \frac{13}{5} \\ &= 3,25,000/- \end{aligned}$$

18). A, B, C enter into partnership. A invest some at the beginning. B invest double the amount of A after 6 months. And C invest thrice the amount of A after 8 months. If the annual profit is 2700. Find the share of C in profit.

Ans: 9000/-

$$P_A : P_B : P_C = 12 \times x : 2x \times 6 : 3x \times 4$$

$$= 12x : 12x : 12x = 1 : 1 : 1$$

$$C's \text{ share} = \frac{1}{3} \times \frac{9000}{2700} = 9000/-$$

$$\begin{array}{r} 12500 \\ \times 13 \\ \hline 37500 \\ 125000 \\ \hline 162500 \\ + 12500 \\ \hline 175000 \end{array}$$